Gyumin Kim

M.S.–Ph.D. Integrated Student,

Department of Electrical Engineering,

Ulsan National Institute of Science and Technology (UNIST)

Contact Information

- Email: q.min@unist.ac.kr
- Lab: Semiconductor Design Automation (SeDA) Lab, UNIST, Ulsan, Republic of Korea

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Education

- *Integrated M.S & Ph.D.*, Dept. of EE, UNIST (2024.9 – Present)
  - Advisor: Prof. Heechun Park
- *B.S.*, Dept. of EE, Kookmin University (2020.3 – 2024.8)

Research Experiences

- *Graduate Researcher*, SeDA Lab, Dept. of EE, UNIST (2024.9 – Present)
  - Advisor: Prof. Heechun Park
- *Visiting Researcher*, Dept. of ECE, University of Southern California, Los Angeles, CA (2025.9 – 2025.10)
  - Two-week research visit for collaboration (Sep 22 – Oct 4, 2025)
  - Host: Prof. Sung Kyu Lim
- *Visiting Researcher*, SeDA Lab, UNIST (2024.3 – 2024.8)
  - Advisor: Prof. Heechun Park
- *Undergraduate Researcher*, SeDA Lab, Kookmin University (2023.9 – 2024.2)
  - Advisor: Prof. Heechun Park

Research Interests

- Physical design methodologies for 3D integration
  - Hybrid-bonded face-to-face (F2F) 3D ICs: tier partitioning and 3D via legalization
  - Unified design flows for homogeneous and heterogeneous 3D ICs
  - Monolithic 3D ICs: thermal-aware placement and cross-tier routability
- Backside design methodologies for advanced technology nodes
  - Backside power delivery network (BS-PDN) optimization for F2F 3D ICs
  - Backside clock distribution network (BS-CDN) synthesis
- Machine learning for electronic design automation (ML-EDA)
  - Reinforcement learning for routing congestion mitigation and 3D via legalization
  - Graph neural networks for physical design optimization
  - Learning-based prediction for placement refinement and thermal analysis

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Teaching Experiences

Fall 2024

- (Undergrad.) Digital Logic and Laboratory (EEE20201, EEE20202) – *Teaching Assistant*

Honors & Awards

- *Best Paper Award*, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD), 2026

- "Segment-Level Rerouting with Reinforcement Learning for Routing Congestion Mitigation"

Publications (accepted and published only)

International Journals (1)

- H. Jung, J. Choi, S. Baek, B. G. Shin, Y. J. Song*, H. Jung, J. Kim, J. Jeon*, **Gyumin Kim**, Heechun Park*, Y. Lee, J. Yoo*, J.-H. Lee, H. Kim, K. Kang*, J. Jeong, S. H. Kim*, J. Bae, C. S. Kim, W. K. Yang, S. Lee, J. Kwon*, B.-S. Kim, J.-H. Han*, H.-J. Kim, H. H. Yoon*, J. Kwon*, and Y. J. Hong*, "Advances and Future Challenges in Monolithic 3D Integrated Logic, Power, and Optoelectronics Technologies for Tightly Interconnected Intelligent Systems", *ACS Nano*, Vol. 20, No. 8, pp. 6407–6445, 2026

International Conferences (8)

- **Gyumin Kim** and Heechun Park, "Power Network Topology Optimization for Top Dies in Backside-Powered F2F 3D ICs", *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2027
- Ikkyum Kim, Junghyun Yoon, **Gyumin Kim**, Jooyeon Jeong, and Heechun Park, "Comprehensive BS-CDN Synthesis Framework with GNN Clustering and MILP-Based Optimization", *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2026
- **Gyumin Kim** and Heechun Park, "Differentiable Thermal-Aware Placement for Monolithic 3D ICs with Stack-Aware Thermal Modeling", *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2026
- **Gyumin Kim** and Heechun Park, "Prediction-Guided Placement Refinement for Cross-Tier Routability in Monolithic 3D ICs", *ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD)*, 2026
- **Gyumin Kim** and Heechun Park, "Segment-Level Rerouting with Reinforcement Learning for Routing Congestion Mitigation", *ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD)*, 2026 **[Best Paper Award]**
- Junghyun Yoon, Ikkyum Kim, **Gyumin Kim**, Sojung Park, and Heechun Park, "Standard Cell Layout Generation: Methodological Evolution and Architectural Impacts", *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2026
- **Gyumin Kim** and Heechun Park, "A Unified Design Flow for Homogeneous and Heterogeneous 3D Integration with Fine-Pitch Hybrid Bonding", *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2025
- **Gyumin Kim** and Heechun Park, "Clustering-Driven Bonding Terminal Legalization with Reinforcement Learning for F2F 3D ICs", *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025

Technical Skills

- EDA Tools: Cadence Innovus, Genus, Celsius; Synopsys ICC2, Design Compiler, PrimeTime, VCS, Verdi
- Programming: Python, C/C++, Tcl, PyTorch
- OS: Linux

(Last updated: Oct. 17, 2026)